

Occupational AI Exposure and Young-Worker Employment: Support and Comparisons in the CPS

Lily Luo

[AUTHOR TO COMPLETE: affiliation]

[AUTHOR TO COMPLETE: email]

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Abstract. Recent evidence associates generative-AI exposure with weaker employment among workers in their early twenties. I reconstruct the public monthly CPS comparison and ask which occupational contrasts support it. The estimand is the post-December-2022 change in the log conditional-mean employment-stock ratio of ages 22–25 to 26–65 in the highest versus lowest GPT-4-exposure quintile, conditional on intermediate quintiles and preperiod software exposure. Although nominal support contains 468 occupations, only four broad families span both tails; their 29 occupations represent 5.03 percent of preperiod stock, and direct-tail information is equivalent to 6.2 occupations. The pooled estimate is -0.132 log points. Broad occupation-family paths reduce it to -0.022 , but the direct-tail interval cannot resolve a within-family gradient. An annual ACS extension narrows person-sampling uncertainty without creating the missing occupational comparisons. Computer-use and family conditioning move the coefficient in opposite directions, preperiod dynamics reject parallel evolution, and linked-CPS estimates do not isolate a flow mechanism. The public-CPS association is supported predominantly by cross-family comparisons; it is not an identified causal AI effect or a replication of employer hiring results.

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1. Introduction

Recent evidence associates generative-AI exposure with weaker employment growth among workers in their early twenties. Interpretation is difficult because exposure is constructed rather than observed, high- and low-exposure jobs differ broadly, and the period around ChatGPT’s release includes several aggregate disruptions.

This paper asks which occupational comparisons support the public Current Population Survey (CPS) pattern. I rebuild monthly employment stocks from January 2017 through July 2026 and classify occupations with the GPT-4 beta score of [Eloundou et al. \(2024\)](#). The target is the post-December-2022 change in the log conditional-mean employment-stock ratio of ages 22–25 to ages 26–65 in the highest relative to the lowest exposure quintile, conditional on intermediate quintiles and preperiod Webb software exposure. I examine broad-family paths, direct within-family support, preperiod computer use, dynamics, and linked-CPS flows. Exposure measures susceptibility rather than adoption; January 2023 is chronology, not an instrument.

Direct support is thin. Four families span both exposure tails; their 29 occupations represent 5.03 percent of preperiod stock, and the interval includes large effects of either sign. The conditioned comparison otherwise relies on intermediate quintiles and common-coefficient restrictions, so its near-zero estimate is not economic equivalence or evidence of no AI effect.

The reconstructed coefficient is -0.132 log points, with 95-percent occupation wild-score interval $[-0.221, -0.044]$; it is neither an individual probability nor an employer hiring rate. With two-digit occupation-family young-relative monthly paths, it is -0.022 $([-0.161, 0.117])$. Their paired difference is 0.110 $([0.011, 0.210])$. This distinguishes conditional estimands without making a causal decomposition: family paths may absorb AI-related and non-AI-related evolution. The coefficient movement is therefore a consequence of the support result, not evidence that one known share of an AI effect is explained by composition.

Annual ACS data separate person-sampling resolution from occupational support. With fixed primary exposure labels, the full-calendar family-year coefficient is negative with an ACS replicate interval excluding zero, but occupation- and family-shock intervals include zero; the nonreuse-calendar family-year coefficient is also undetected. More respondents therefore narrow one uncertainty component without creating the missing within-family tail comparisons.

Adding a pre-2022 O*NET computer-use measure makes the common-support coeffi-

cient more negative; one control therefore does not resolve computerization. Unrestricted quarterly coefficients reject preperiod equality, while linked-flow intervals include zero. The data neither isolate an AI counterfactual nor attribute the stock pattern to exit, switching, or entry allocation.

The paper shows that a sizeable public-CPS association is identified predominantly by comparisons spanning broad occupation families, quantifies the resolution of direct and flow evidence, and records calendar and taxonomy failures that silently alter support. This localization tells future adoption or quasi-experimental studies which comparisons they must replace rather than inherit. It narrows rather than overturns the motivating evidence: it neither replicates proprietary payroll outcomes nor identifies AI, computerization, or pandemic recovery as the cause.

2. Related Evidence and the Empirical Object

The closest motivating study is the August 2026 revision of [Brynjolfsson, Chandar, and Chen \(2026\)](#). Using ADP payroll records through June 2026, it reports weaker employment growth among ages 22–25 in occupations with high GPT-4 beta exposure and studies employer-match hiring and separation. The CPS cannot reproduce its employer panel, standardized-title mapping, firm controls, or worker–firm flows. Because that study also reports public CPS and ACS evidence, novelty here lies in auditing a fixed national-CPS estimand, not discovering the age pattern.

Administrative and workplace studies observe adoption or tool access in particular settings ([Humlum and Vestergaard 2025](#); [Brynjolfsson, Li, and Raymond 2025](#)). Firm-level evidence also distinguishes direct task substitution, reallocation across tasks, and firm-growth effects ([Hampole et al. 2025](#)). Survey evidence shows that realized generative-AI use varies substantially even among workers doing similar work and that exposure explains only part of adoption ([Bick et al. 2026](#)). Occupation scores instead describe tasks a technology might affect. Remote-work evidence supplies another competing interpretation ([Emanuel, Harrington, and Pallais 2026](#)). None turns a capability score into observed deployment.

The primary score is Eloundou et al.’s GPT-4 beta: tasks are labeled by whether an LLM, alone or with complementary software, could reduce completion time at constant quality ([Eloundou et al. 2024](#)). AIOE maps AI applications to abilities ([Felten, Raj, and Seamans 2018, 2021](#)); Webb measures patent–task overlap ([Webb 2020](#)); Dingel–Neiman classifies teleworkability ([Dingel and Neiman 2020](#)). Computer use, routine content, education, and

remotability are therefore overlapping occupational characteristics, not clean alternative treatments. The task literature likewise allows technology to substitute for some tasks, complement others, and reallocate work within occupations (Autor, Levy, and Murnane 2003; Autor and Dorn 2013; Acemoglu and Restrepo 2019). Faster-changing technical careers may also combine early entry advantages with skill obsolescence later in the career (Deming and Noray 2020), while the employment consequence of automating a task depends on whether the remaining bundle requires more or less expertise (Autor and Thompson 2025). These mechanisms make an ages-22–25 versus older-worker contrast economically relevant but not uniquely diagnostic of entry-level AI substitution.

This distinction changes the burden of interpretation. Agreement across correlated scores can show that a direction is not unique to one coding rule, but cannot validate a latent exposure construct. Conversely, coefficient differences can arise from task definitions, occupational support, or treatment-group membership. The empirical work below therefore matches supports where possible and names population changes where it is not.

I distinguish six layers:

$$C \longrightarrow M \longrightarrow H \longrightarrow \Omega \longrightarrow R \longrightarrow Q,$$

the intended construct, measurement rule, occupational harmonization, retained support, representation, and regression estimand. A vintage mismatch is an implementation error; score repair on fixed support changes measurement; occupation readmission changes population; and a different exposure contrast changes the estimand. Common-draw paired inference tests coefficient movement, while each component interval measures its own resolution. Neither a detected movement nor a nondetection assigns a causal share or establishes equivalence. This focus connects the exercise to specification-sensitivity and generated-covariate inference (Andrews, Gentzkow, and Shapiro 2017; Rambachan and Roth 2023; Battaglia et al. 2024; Christensen and Hansen 2026; Duan and Pelger 2026; Ludwig, Mullainathan, and Rambachan 2026).

3. Data and Exposure Construction

3.1. CPS stocks and calendar

I use IPUMS Basic Monthly CPS data from January 2017 through July 2026. Employed respondents ages 18–65 with positive final weights are aggregated to occupation–month stocks; the regression compares ages 22–25 with ages 26–65. Age is not experience, and

older workers need not be untreated.

The corrected source calendar contains 114 of 115 possible months. Five March files in the wide extract were ASEC samples and are replaced by authenticated Basic Monthly files for 2017–2021. October 2025 is absent because no CPS was collected during the funding lapse; it is not interpolated. December 2022 is observed but excluded as the transition month, leaving 113 static months.

The reconstruction scans 9,843,021 source rows and retains 5,322,047 eligible records before routing. Fractional pre-2020 crosswalk descendants are not distinct respondents. The 468-occupation grid contains 52,884 cells; 51,891 enter the likelihood, including 12,965 valid one-sided zeros, while 993 both-age-zero cells contribute nothing. Young cells are sparse: 26.25 percent have zero respondent-equivalent records and 69.83 percent have fewer than five.

TABLE 1. Corrected CPS reconstruction and analysis support

	Count	Definition
Expected calendar months	115	Jan. 2017–July 2026
Observed source months	114	October 2025 absent
Static analysis months	113	December 2022 excluded
Eligible employed source records	5,322,047	ages 18–65, positive final weight
Route-expanded rows	6,188,956	fractional descendants allowed
Primary occupations	468	finite beta and Webb; positive pre-stock
Occupation–month grid	52,884	468 occupations × 113 months
Positive-total fitted cells	51,891	one-sided zeros retained
Both-age-zero cells	993	no likelihood contribution

Notes: Source records are not independent observations for regression inference. Before 2020, one source record can generate several fractional crosswalk descendants. Counts, routing checks, named support, exclusions, and restricted-input hashes are generated by the corrected-calendar data audit.

3.2. Harmonization and support

Pre-2020 Census-2010 occupations are routed to Census-2018 using versioned official shares. Applying the same shares to young and older records preserves source stock and age ratios by construction, but does not validate age-specific routing. The route reaches 99.546 percent of eligible early stock and conserves routed weight to machine precision.

Primary support requires finite GPT-4 beta and Webb software scores and positive preperiod young and older stock. It contains 468 occupations and 87.19 percent of

candidate-universe stock; a named file records all 73 exclusions. Raw AIOE is a standardized source-occupation index, not a percentage; the source workbook has mean approximately zero, unit sample standard deviation, and an interquartile difference of 1.879 raw points. A literal SOC-2010 AIOE merge into post-2018 employment data covers only 3.33 percent of computer-and-mathematical employment, versus 97.7 percent after version-aware routing. Fixed-support score repair changes little; occupation readmission changes the population. I treat the literal merge as invalid and do not attribute it to another paper without implementation evidence.

The support file separates missing exposure, missing Webb, and absent young preperiod stock. This matters because a common-support regression estimates a population defined by all required scores, whereas native-support rows can change both the exposure definition and who remains. The appendix's four-step crosswalk table holds support fixed before adding newly reachable occupations, so measurement correction is not conflated with sample composition.

3.3. Exposure groups and production breaks

The primary exposure is Eloundou et al.'s GPT-4 beta share. CPS stock from January 2017 through November 2022 sets normalization and employment-weighted quintile cuts, with ties retained. The quintiles contain 129, 99, 82, 67, and 91 occupations and about one fifth of preperiod stock each. No postperiod stock enters the rebuilt labels.

Employment weighting balances stock rather than occupation counts. Ties are assigned to the lower group at a cut and never split to force exact fifths. Consequently, group membership is a versioned treatment object rather than a percentile transformation that can be recreated from rounded published values.

A historical production treatment instead used the full static panel for weights. On the corrected calendar its coefficient is -0.1346 . Rebuilding from the 71-month preperiod changes nine assignments and moves the coefficient by 0.0024 log points ($[-0.0013, 0.0062]$). The revised analysis uses only the rebuilt treatment; the historical row is a checkpoint.

Official documentation also identifies population-control changes in January 2025 and January 2026, modified November 2025 collection and weighting, and later longitudinal-identifier corrections. These establish survey-production breaks, not their causal effect on the coefficient. I report endpoint and shutdown-month sensitivities, preserve elapsed calendar time in lags, and do not apply aggregate population adjustments to age-by-occupation cells without the required counterfactual controls.

4. Estimand and Inference

Let N_{oat} be CPS-weighted employment stock in occupation o , age group a , and month t ; Y_a indicate ages 22–25; P_t indicate January 2023 onward; Q_{oq} indicate rebuilt exposure quintile q ; and W_o be standardized Webb software exposure. The pooled mean is

$$E[N_{oat} | X] = \exp \left[\gamma_{oa} + \delta_{ot} + \lambda_{at} + \sum_{q=2}^5 \beta_q Q_{oq} Y_a P_t + \theta W_o Y_a P_t \right]. \quad (1)$$

The target β_5 is the postperiod change in the young-to-older stock ratio for Q5 relative to Q1, conditional on Q2–Q4 and Webb. It is not an individual probability, continuous slope, or hiring rate.

With two age groups, conditioning on $T_{ot} = N_{oyt} + N_{oot}$ removes the occupation–month effect and gives

$$\ell_{ot} = N_{oyt} \log p_{ot} + N_{oot} \log(1 - p_{ot}), \quad \text{logit}(p_{ot}) = \log(\mu_{oyt}/\mu_{oot}). \quad (2)$$

Final weights construct continuous cell stocks once; equation (2) is an estimating criterion, not a literal independent-binomial sampling model.

The central alternative gives each SOC two-digit family its own young-relative calendar-month path; a less saturated companion uses family-specific post shifts. Both change the conditional estimand. Neither the residual coefficient nor its movement from equation (1) is an AI remainder or composition share.

The occupational-conditioning, influence, dynamic, and flow analyses added in this revision were designed after the initial outcome pattern was known. They are post-outcome exploratory analyses, not preregistered confirmatory tests; simultaneous inference does not undo that selection.

4.1. What the design establishes

The design measures the magnitude, support, and specification sensitivity of a public-CPS association. It distinguishes paired coefficients and locates the occupational comparisons carrying the contrast, but cannot allocate movement among AI, computerization, pandemic recovery, or other forces. Attenuation is not a causal share, survival is not an AI remainder, and an interval containing zero is nondetection rather than economic equivalence. The limitations section returns to preperiod evolution, sparse direct support, exposure without adoption, and incomplete flow and survey-design information.

For shared observations I reuse bootstrap multipliers and form each draw of $\Delta = \widehat{\beta}^{(2)} - \widehat{\beta}^{(1)}$. A paired interval excluding zero shows statistical distinguishability under that procedure; it does not identify the reason. An interval including zero means nondetection, not economic equivalence. I report $\text{MDE}_{80} \simeq (1.96 + 0.84)SE$ only as a precision diagnostic.

To describe where a target is learned, let $h_{ot} = T_{ot}p_{ot}(1 - p_{ot})$ and let r_{ot} be the target regressor after projection on every nuisance regressor in the fitted-information metric. I report

$$I = \sum_{o,t} h_{ot} r_{ot}^2, \quad s_o = \frac{\sum_t h_{ot} r_{ot}^2}{I}, \quad G_{\text{eff}} = \left[\sum_o s_o^2 \right]^{-1}.$$

These are fitted target information and concentration diagnostics, not an effective number of CPS sampling clusters. Family and occupation shares, leave-one-out estimates, and the top-five information share are reported without using influence to select a preferred sample.

The principal 9,999-draw wild-score interval treats occupations as shock clusters and allows arbitrary serial dependence within occupation. It conditions on weighted cells, exposure labels, mapping, support, and specification. Separate sensitivities use 22 broad-family shocks, household/sample-unit full refits under released weights, and elapsed-calendar time-HAC scores. A sparse-cell and family-shock simulation audits finite-sample behavior. These procedures have different stochastic targets; they are not mechanically combined, and none supplies complete CPS design inference or exposure-label uncertainty.

The household exercise keeps every month and fractional route descendant of a positive CPS household identifier together. It still omits unavailable multistage-selection and calibration uncertainty. The time-HAC construction uses elapsed calendar lags, so the missing October 2025 is not treated as adjacency between September and November.

5. Which Occupational Comparisons Support the Pattern?

5.1. Public benchmark reconstruction

We first reconstruct the public stock comparison before conditioning on broad occupation families. These are independent CPS analogues using equal-occupation Rule-A beta quintiles on the fixed primary support, not an exact replication of BCC: exhaustive BCC occupation membership is unavailable. For all employed workers, the annual 2022–2024 Q5–Q1 growth-factor difference is -0.0346 ($[-0.1455, 0.0763]$); the November 2022–June 2026 contrast is -0.2312 ($[-0.4660, 0.0036]$). The top-two/bottom-three dashboard analogue is a dif-

ferent grouping, with estimates -0.0269 ($[-0.0972, 0.0435]$) and -0.1007 ($[-0.2380, 0.0367]$), respectively. Its inference is computed separately; no Q5–Q1 rejection or precision statement transfers to it.

No-control occupation long differences provide another distinct estimand. With baseline employment-stock weights and HC1 standard errors, the all-employed Q5–Q1 coefficient is -0.0318 ($[-0.1447, 0.0811]$) over 2022–2024 and -0.2466 ($[-0.4771, -0.0160]$) from November 2022 to June 2026. On the 468-occupation common support, the separately defined no-Webb young-relative model gives a pooled coefficient of -0.1146 ($[-0.2057, -0.0236]$); allowing family-specific young-relative calendar-month paths gives -0.0124 ($[-0.1446, 0.1197]$). Family conditioning therefore changes the descriptive estimand by absorbing cross-family evolution. It does not identify a causal AI effect, and intervals containing zero establish non-detection rather than equivalence.

An independent annual ACS reconstruction gives a different view of sampling precision. Between 2022 and 2024, the equal-occupation BCC analogue has an all-employed Q5–Q1 growth-factor difference of -0.0859 ($[-0.1238, -0.0480]$ under ACS successive-difference replication) and a full-time civilian wage-and-salary difference of -0.0834 ($[-0.1272, -0.0396]$). The corresponding published BCC comparisons are -0.022 and -0.019 . Because exhaustive BCC occupation membership is not public, the gap cannot be assigned separately to samples and grouping rules.

The annual young-relative ACS extension also remains sensitive to broad-family conditioning and calendar construction. On the fixed primary grouping and the full 2017–2019, 2021, 2023–2024 calendar, the all-employed coefficient moves from -0.1089 ($[-0.1310, -0.0869]$) pooled to -0.0474 ($[-0.0885, -0.0063]$) with family-year effects. On a nonreuse calendar that drops 2018–2019, the corresponding estimates are -0.0951 ($[-0.1204, -0.0698]$) and -0.0220 ($[-0.0705, 0.0264]$). These intervals capture ACS person-sampling uncertainty under a year-block approximation. Occupation- and family-shock intervals are wider and include zero for each of the four reported family-year variants. The ACS therefore supplies a larger public-sample analogue, not an exact payroll replication or causal AI design.

Table 2 begins with the rebuilt pooled result. The Q5–Q1 coefficient is -0.1321 , with standard error 0.0452 clustered by occupation. Its 95-percent interval is $[-0.2206, -0.0437]$. Giving every SOC two-digit family its own young-relative calendar-month path yields -0.0217 ($[-0.1607, 0.1173]$). On common draws, the conditioned-minus-pooled movement is 0.1104 ($[0.0107, 0.2102]$), with paired $MDE_{80} = 0.1454$. Thus the conditional estimands are distinguishable while the residual gradient is imprecise.

TABLE 2. Pooled and broad-family exposure comparisons

Specification	Coefficient	95% interval	Paired movement	MDE ₈₀
Pooled Q5–Q1	–0.1321	[–0.2206, –0.0437]	—	0.1266
SOC2 × young × month	–0.0217	[–0.1607, 0.1173]	0.1104	0.1998

Notes: The weighted-stock criterion is the grouped-binomial conditional likelihood for the two age-specific employment-stock counts. Coefficients parameterize log ratios of conditional means; no observed log ratio is constructed, and one-sided zero-stock cells remain in the criterion. The second row changes the conditioning estimand. Intervals use 9,999 occupation wild-score draws. The paired movement is the second-row coefficient minus the first on common draws; its interval is [0.0107, 0.2102] and its MDE₈₀ is 0.1454. MDEs are normal-theory precision diagnostics, not rejection thresholds.

The ratio 0.1104/0.1321 is not a causal “share explained.” Broad families differ in AI-relevant tasks, computerization, education, work location, and pandemic recovery; family paths absorb all such common evolution. The result says that the pooled estimate relies mainly on comparisons across families, not what would have happened without AI.

The full Q2–Q5 profile reaches the same conclusion. Pooled coefficients are jointly different from zero but do not establish a monotone dose response. After family-month conditioning, the four coefficients are jointly indistinguishable from zero. Simultaneous intervals and paired movements appear in the online appendix.

5.2. Direct support and restrictions

Only SOC27, 29, 31, and 41 contain Q1 and Q5 occupations. Elsewhere the regression links partial segments—for example Q1–Q3 in one family and Q4–Q5 in another—through common quintile coefficients. That is valid regression variation but not repeated side-by-side tail comparisons.

A direct-tail benchmark uses only Q1 and Q5 occupations in the four spanning families. Its 29 occupations account for 5.03 percent of preperiod stock. The family-month estimate is 0.1494 ([–0.1694, 0.4682]), so the CPS cannot precisely resolve a direct tail contrast within broad families. This row changes population and support and cannot replace the primary estimand.

A continuous companion instead subtracts each family’s preperiod-weighted beta mean and imposes a common within-family slope. Its family-month coefficient is –0.0025 per 0.1084 raw beta units ([–0.0240, 0.0191]). It uses more residual variation but trades the tail contrast for linearity and a common slope. Together, the specifications expose the support–functional-form tradeoff.

The pre-2020 occupational bridge adds a separate allocation uncertainty. A jointly fea-

TABLE 3. Exposure-quintile support within broad occupation families

SOC2	Occupation family	Q1	Q2	Q3	Q4	Q5	Both tails
11	Management	0/0.00	0/0.00	15/4.85	12/8.45	2/0.25	
13	Business and financial	0/0.00	0/0.00	1/0.74	16/2.42	11/3.06	
15	Computer and mathematical	0/0.00	0/0.00	0/0.00	1/0.03	14/4.29	
17	Architecture and engineering	0/0.00	0/0.00	3/0.12	9/1.49	7/0.67	
19	Life, physical, and social science	0/0.00	0/0.00	12/0.86	6/0.34	3/0.05	
21	Community and social service	1/0.03	5/0.59	6/0.69	0/0.00	0/0.00	
23	Legal	0/0.00	0/0.00	1/0.91	2/0.35	1/0.08	
25	Education and library	0/0.00	3/3.52	2/0.85	0/0.00	2/0.16	
27	Arts, media, and sports	2/0.03	5/0.27	3/0.27	6/0.57	9/0.65	Yes
29	Healthcare practitioners	3/0.20	18/1.94	11/3.80	3/0.21	2/0.23	Yes
31	Healthcare support	4/1.58	6/1.45	3/0.61	0/0.00	1/0.03	Yes
33	Protective service	3/0.30	8/1.05	6/0.92	0/0.00	0/0.00	
35	Food preparation and serving	5/1.14	4/1.76	0/0.00	1/0.37	0/0.00	
37	Building and grounds maintenance	3/1.96	2/0.34	1/0.20	0/0.00	0/0.00	
39	Personal care and service	7/1.72	8/0.56	2/0.23	0/0.00	0/0.00	
41	Sales and related	1/0.03	2/1.53	3/2.75	4/3.48	7/2.28	Yes
43	Office and administrative support	0/0.00	5/0.80	3/0.04	7/2.60	32/7.77	
45	Farming, fishing, and forestry	3/0.09	2/0.05	0/0.00	0/0.00	0/0.00	
47	Construction and extraction	27/5.49	1/0.02	2/0.62	0/0.00	0/0.00	
49	Installation, maintenance, and repair	21/2.65	8/0.56	2/0.35	0/0.00	0/0.00	
51	Production	39/2.13	9/0.72	4/1.28	0/0.00	0/0.00	
53	Transportation and material moving	10/2.69	13/4.77	2/0.06	0/0.00	0/0.00	

Notes: Each Q cell is occupation count/national preperiod employment-stock share in percent. The producer forms each occupation's stock by summing monthly CPS-weighted young and older employment over the 71 observed months from January 2017 through November 2022. Because every occupation contributes the same months, these shares also equal shares based on average-month stock. They sum to 100 percent across the 110 cells; zero cells are shown rather than suppressed. The matrix uses the fixed 468-occupation support. Only four families contain both Q1 and Q5. The pooled regression connects other families through intermediate quintiles and common coefficients, but they do not supply direct within-family tail comparisons. Family totals, beta ranges, and information shares are in Online Appendix Table C.1.

sible odds-tilt grid reallocates the 14.86 percent of supported early-period stock routed through 31 split source occupations while preserving source-age-month mass. With exposure groups rebuilt after each allocation, the pooled coefficient ranges from -0.1446 to -0.1163 , and all six paired intervals relative to the no-tilt allocation contain zero. The family-month estimand is less stable: at the $K = 0.25$ endpoint it moves from -0.0217 to 0.0439 , a paired change of 0.0656 ($[0.0031, 0.1280]$), and five of six nonbaseline paired intervals exclude zero. A wider 25-point feasible grid places the pooled coefficient between -0.1599 and -0.0702 . These are post-outcome mapping sensitivities over finite declared grids, not estimates of true age-specific routes, sharp bounds, or identified sets.

5.3. Influence and observed paths

The online appendix reports nuisance-adjusted information, effective information counts, and family and occupation deletions. Fast-food and counter workers have substantial signed influence because low-exposure service employment recovered differently after the pandemic, but other occupations push oppositely. Outcome-selected deletion is not a design correction; symmetric trimming and economically defined service exclusions remain diagnostics.

Figure 1 separates young and older stocks for both tails. Exact equal-month accounting gives a Q5–Q1 young-stock change of -0.1098 log point and an older-stock change of 0.0483 , so their young-relative difference is -0.1581 . The numerator supplies the larger movement, while the denominator reinforces it. This identity is not the grouped-binomial coefficient -0.1321 , and neither component is an individual probability or a causal share. A symmetric family accounting assigns -0.1293 to within-family ratio changes and -0.0288 to changing older-family weights; the appendix reports the zero-denominator audit. The low-exposure reference trajectory is part of every comparison.

6. Characteristics and Inferential Resolution

The leading computerization measure is the May 2020 O*NET rating “Interacting With Computers.” On the 455-occupation matched support, its preperiod-weighted correlation with beta is 0.797 . Table 4 holds that support, treatment, outcome, and calendar fixed while crossing a computer-use control with family-specific young-relative monthly paths. The pooled coefficient is -0.0958 . Adding computer use moves it to -0.1966 ; the paired change is -0.1009 ($[-0.1772, -0.0245]$). Adding family paths without computer use instead gives 0.0354 , while adding both gives -0.0467 . The combined-minus-baseline change is 0.0491

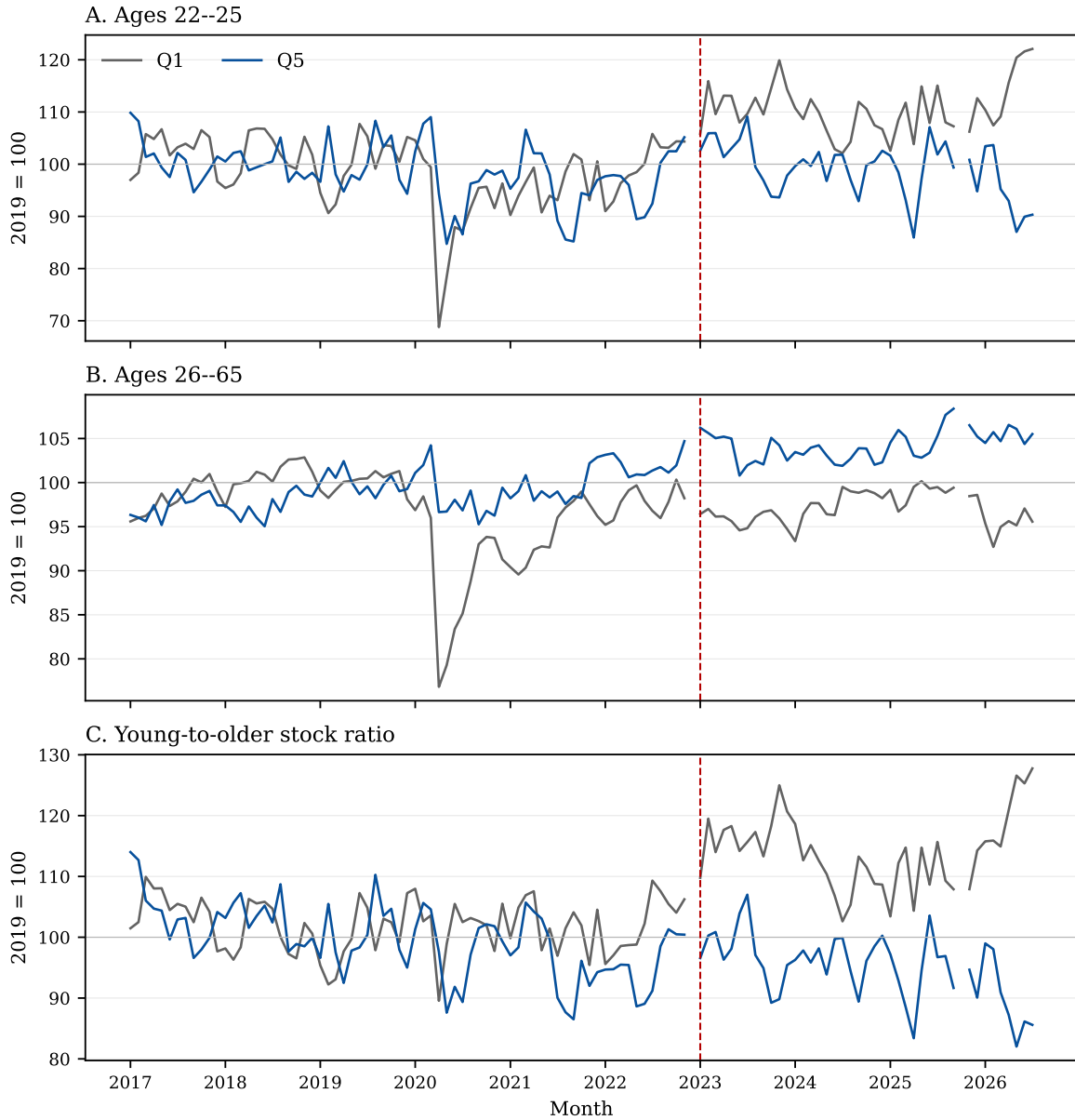


FIGURE 1. Young and older employment stocks in the exposure tails

Notes: Monthly CPS-weighted stocks use rebuilt beta labels and are indexed to each series' 2019 mean. The ratio panel divides young by older stock before indexing. The vertical line marks January 2023; December 2022 is the transition month, and October 2025 is absent rather than interpolated. Paths are descriptive.

($[-0.0642, 0.1624]$).

The computer-use-by-young-by-post coefficient is 0.0460 per preperiod- weighted standard deviation in the pooled computer model ($[0.0117, 0.0803]$) and 0.0487 in the combined model ($[0.0105, 0.0868]$). Computer use reduces retained Q5 target information from 69.1 to 29.0 percent in the pooled comparison, a VIF-like ratio of 3.45. The positive computer coefficient and more negative exposure coefficient are joint-projection facts under collinearity, not evidence that computerization raises employment or that the residual isolates AI.

On a separate 347-occupation common support, remotability, wages, education requirements, and routine/manual content do not individually produce detected movements from their common baseline; computer use remains the exception. A repaired pandemic-shortfall exercise uses bounded predictions and a stricter 363-occupation support; total-stock and young-relative conditioning move the pooled coefficient by only 0.0004 and 0.0011 log point. Regenerating these controls within linked-household draws widens uncertainty but does not resolve a conditioning movement. These are exploratory diagnostics, not equivalence results.

TABLE 4. Computer-use and broad-family conditioning on common support

Family-month paths	Computer-use control	Occupations	Q5 by post [95% CI]	Change from pooled baseline [95% CI]
No	No	455	-0.0958 $[-0.1807, -0.0108]$	—
No	Yes	455	-0.1966 $[-0.3161, -0.0771]$	-0.1009 $[-0.1772, -0.0245]$
Yes	No	455	0.0354 $[-0.0859, 0.1567]$	0.1311 $[0.0352, 0.2271]$
Yes	Yes	455	-0.0467 $[-0.1821, 0.0888]$	0.0491 $[-0.0642, 0.1624]$

Notes: All four post-outcome exploratory rows use the same 455 occupations, rebuilt exposure labels, outcome cells, calendar, and 9,999 common occupation multipliers. The computer-use rating is standardized with preperiod employment weights and interacted with young $\times$ post. Family-month rows give every SOC2 family its own young-relative calendar-month path. The final column uses covariance-preserving paired draws. The four projections are not additive components, and no residual coefficient is a causal AI effect.

Broad-family paths and a computer-use control need not move the coefficient alike: the former remove all family-common evolution, while the latter partials one correlated dimension. The matched four-model comparison shows that their projections interact rather than add. Neither is a causal technology decomposition.

Industry conditioning is built from occupation–industry–age–month cells rather than dominant-industry labels. Allowing 13 broad industries their own young-relative post shifts moves the stratified coefficient from -0.138 to -0.099 ; the paired 0.039 interval $[0.008, 0.070]$ excludes zero. It does not measure firm AI adoption. BA and non-BA estimates are -0.138 and -0.123 ; their paired difference is -0.015 ($[-0.156, 0.125]$), with $MDE_{80} = 0.201$. Exact-age estimates vary but simultaneous paired intervals do not isolate

a particular age mechanism.

The industry row changes the cell objective as well as the nuisance set, so it is reported separately from occupation-level characteristic controls. Education and exact-age estimates use literal common supports and common draws; no subgroup is selected by its point estimate. The age coefficients are jointly unequal, but each simultaneous age-minus-pooled interval contains zero.

Changing the older comparison band does not remove the pooled pattern. For ages 22–25 relative to ages 26–65, 26–30, 31–40, 41–50, and 51–65, the pooled coefficients are respectively -0.132 , -0.121 , -0.160 , -0.139 , and -0.097 . Relative to the 26–65 benchmark, only the 51–65 comparison has a paired interval excluding zero: the movement is 0.035 ($[0.004, 0.066]$). All five family-month coefficient intervals include zero. Holding the older group’s preperiod exact-age composition fixed changes the pooled coefficient by only 0.0008 ($[-0.0011, 0.0027]$). Restricting the young numerator to nonenrolled workers changes it from -0.132 to -0.124 , a paired movement of 0.008 ($[-0.012, 0.029]$). These are non-detections, not evidence of equivalence, and older workers are comparison groups rather than untreated controls.

Separate annual profiles show heterogeneous, nonmonotone enrollment and education changes across the five exposure quintiles. The national ages-22–25 profile, reported without assigning exposure to nonworkers, also changes in enrollment, education, and weighted population, so these diagnostics document rather than eliminate cohort composition as an alternative (Online Appendix D.6).

Table 5 separates uncertainty targets. Occupation wild scores allow serial dependence within occupation. A 22-family sensitivity permits common broad-family shocks. Household full-refit multipliers preserve repeated observations and co-residence under released weights but are not full CPS design inference. None covers exposure labels or route allocation.

With 22 family clusters, the paired movement remains detected under Rademacher and Webb weights; the Webb interval is $[0.0059, 0.2150]$. The completed finite-sample audit finds small coefficient bias but target-specific interval performance. In the empirical observed-effect design, occupation-wild coverage is 0.865 for the pooled coefficient, 0.921 for the family-conditioned coefficient, and 0.889 for their paired movement; family clustering is conservative for the pooled and paired targets but covers the family-conditioned target only 0.895 of the time. In the clean zero-target design, rejection rates for the family-conditioned coefficient are 6.1 percent under occupation weights, 8.9 percent under family weights, and 5.6 percent for the cross-fit oracle. No one procedure is therefore validated

TABLE 5. Inference sensitivities for the core occupational comparison

Procedure and target	Pooled 95% CI	Conditioned 95% CI	Paired-movement 95% CI
<i>Same pooled/SOC2-by-calendar-month targets</i>			
Occupation wild score, SOC2-by-month	[-0.221, -0.044]	[-0.161, 0.117]	[0.011, 0.210]
Twenty-two-family Webb wild score	[-0.211, -0.053]	[-0.153, 0.110]	[0.006, 0.215]
<i>Sampling-oriented companion with regenerated preperiod labels</i>			
Household full refit, 399 draws	[-0.186, -0.076]	[-0.124, 0.066]	[0.035, 0.177]

Notes: All rows report the reconstructed pooled, SOC2-by-calendar-month, and paired-movement coefficients. The 22-family row permits common shocks within broad occupational families. The household row uses 399 mean-one linked-household full refits and regenerates preperiod exposure labels; it is conditional on released CPS final weights and is not complete CPS design inference. The procedures target different perturbations and are not mechanically combined.

for all targets and declared designs. In the rebuilt-treatment audit, elapsed-calendar HAC leaves scalar uncertainty similar and none of the 15 five-parameter covariance matrices across three targets and five lags is indefinite. Earlier indefinite matrices belong to the superseded historical-treatment audit and remain labeled in the archive.

7. Dynamics, Flows, and External Comparison

7.1. Dynamics

The binary post coefficient is not an event-study design. I estimate quarterly coefficients for all four nonreference quintiles, omitting 2022Q4 (October–November; December is the transition month). The Q5 path has 23 preperiod and 15 postperiod coefficients in pooled and family-month specifications.

Figure 2 shows the rebuilt Q5 path. Relative to the omitted 2022Q4 coefficient, joint tests reject that all 23 preperiod coefficients are zero in the pooled ($p = .0171$) and family-conditioned ($p = .0053$) models. A reference-invariant test of equality within the entire preperiod rejects for the pooled path ($p = .0121$) but not for the family-conditioned path ($p = .0641$). The appendix reports fixed pre-pandemic, post-reopening, and pandemic-exclusion windows. An insignificant anchored linear slope cannot override the unrestricted pooled test, while the family result illustrates the importance of the unusually short 2022Q4 reference. Neither pattern supports a causal January-2023 interpretation. When 2020Q2–2020Q4 are excluded, the pooled test no longer rejects at five percent under either the short-reference null ($p = .0772$) or the reference-invariant within-window null ($p = .0676$). For the family-conditioned path the corresponding values are $p = .0079$ and

$p = .1048$, underscoring that the conclusion depends on the null as well as the calendar window.

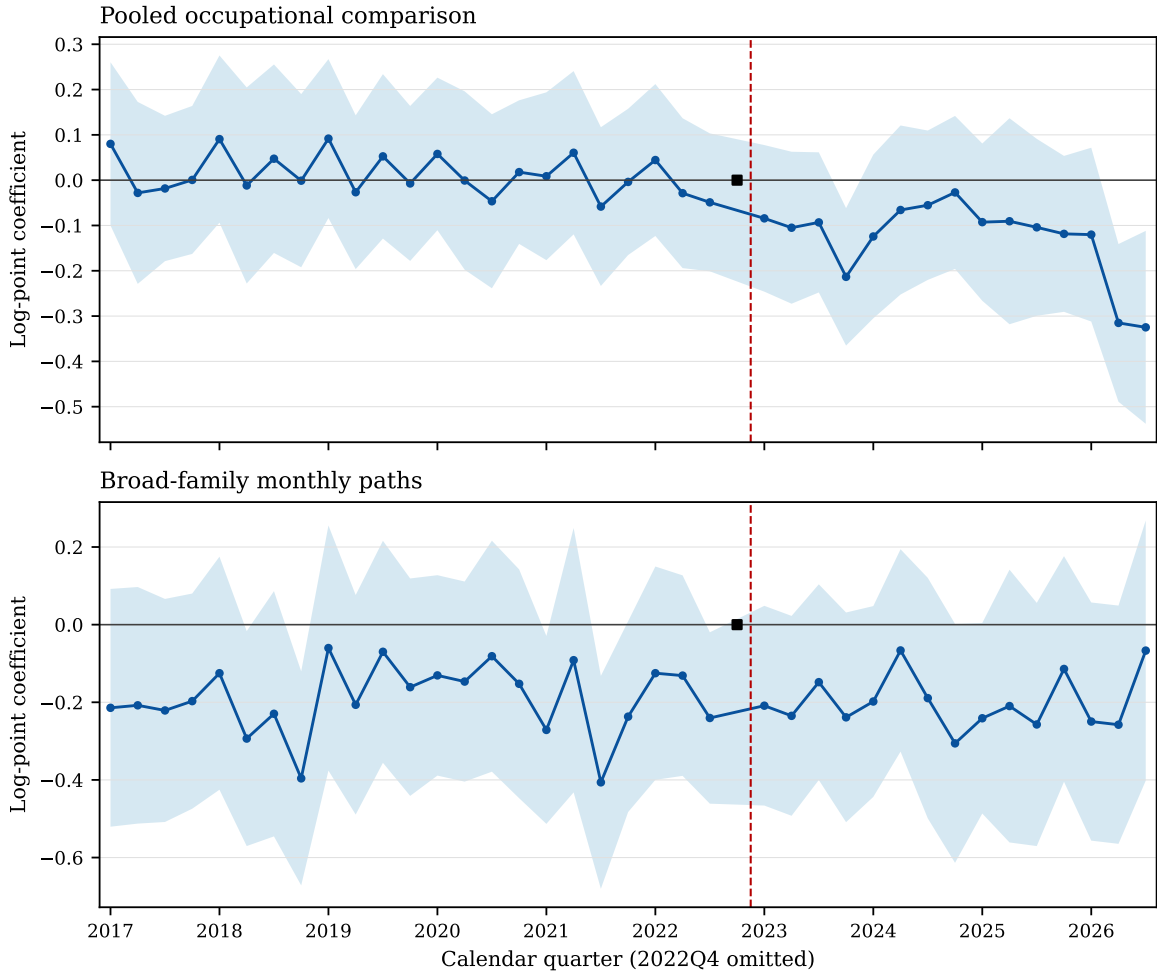


FIGURE 2. Quarterly Q5–Q1 young-relative employment coefficients

Notes: Coefficients are relative to Q1 and omitted 2022Q4 under rebuilt labels; lines are pointwise 95-percent occupation wild-score intervals. The online appendix reports simultaneous bands and the Q2–Q5 system. The conditioned model gives each SOC2 family its own young-relative monthly path.

Three related targets separate a genuine estimand comparison from an event-time normalization artifact. The static grouped-binomial coefficient S is -0.1321 pooled and -0.0217 after family conditioning. The calendar-weighted post functional relative to omitted 2022Q4, P , is -0.1199 ($[-0.2637, 0.0239]$) and -0.2074 ($[-0.4252, 0.0103]$). By contrast, the reference-invariant post-minus-pre functional D is -0.1314 and -0.0217 . The paired $D - S$ differences are 0.0007 ($p = .454$) and less than 0.0001 in absolute value ($p = .995$). Thus D nearly reproduces the static estimates; the design does not detect the family-model $P - S$ difference at five percent (-0.1858 , $p = .074$). What it does detect is that family condition-

ing changes $P - S$ by -0.1980 ($p = .021$), while it does not detect a change in $D - S$ (-0.0008 , $p = .693$). The short-reference functional therefore responds to conditioning differently from the reference-invariant comparison. This numerical reconciliation is not an equivalence claim and does not make any estimand causal.

Official HonestDiD 0.2.8 applies smoothness and relative-magnitude restrictions to the full event vector and covariance for P , not for S or D . Both the full 2017Q1–2022Q3 calibration and a consecutive 2021Q1–2022Q3 post-reopening calibration are reported. Under the latter, the family-conditioned smoothness interval is $[-0.772, -0.007]$ when differential-trend bias is restricted to be exactly linear ($M = 0$), but expands to $[-1.100, 0.207]$ at the first positive grid value, $M = 0.005$. Both conventional P intervals already include zero, so no positive zero-exclusion threshold exists. The exercise cannot repair rejected preperiod equality or identify adoption.

Onset estimates from November 2022 through June 2023 are similar within each structure, but common timing does not distinguish AI from coincident shocks. Ending in December 2024 gives a pooled coefficient of -0.111 ($[-0.201, -0.021]$); its paired movement relative to the July-2026 estimate is 0.021 ($[-0.007, 0.049]$). The corresponding movements at the September- and December-2025 endpoints are 0.0225 and 0.0212 , essentially the same magnitude despite different interval endpoints. The full window remains the canonical target and December 2024 a prominent pre-production-break benchmark; neither is causally clean. Splitting the postperiod instead gives -0.111 in 2023–24 and -0.168 in 2025–26; their paired difference is -0.057 ($[-0.125, 0.012]$), so the official-weight design does not detect a change between eras. Family-conditioned endpoint changes remain near zero, and survey-gap exclusions are small. A coding-stable post-2020 fit gives -0.118 pooled and -0.030 with family-month effects; their paired movements from the full-window models are 0.014 ($[-0.029, 0.057]$) and -0.009 ($[-0.105, 0.088]$). Quintile- and occupation-specific month-of-year controls move either coefficient by at most 0.0011 . All six replacement fits pass the same numerical certification; the earlier failed attempts remain in the failure registry rather than being treated as results.

7.2. Linked CPS margins

Table 6 estimates distinct adjacent- and twelve-month margins using validated links and official longitudinal weights. Every primary interval includes zero. Normal-theory MDE_{80} values range from 0.115 to 0.336 log point, so these are low-resolution nondetections rather than evidence of absent flow responses.

Employment exit starts from employed origins; reported occupation change requires

TABLE 6. Corrected CPS flow margins

Endpoint and margin	Raw link records at risk / entries	Coefficient	95% interval	Normal MDE ₈₀
Adjacent employment exit	3,346,227	0.132	[-0.042, 0.307]	0.254
Adjacent reported-occupation change	3,207,598	0.003	[-0.118, 0.123]	0.174
Adjacent entry-destination allocation	96,981	-0.070	[-0.250, 0.110]	0.260
Twelve-month employment exit	1,399,376	0.123	[-0.111, 0.357]	0.336
Twelve-month reported-occupation endpoint change	1,110,024	-0.018	[-0.097, 0.061]	0.115
Twelve-month entry-destination allocation	82,573	-0.052	[-0.251, 0.147]	0.282

Notes: Counts are repeated endpoint-link records, not unique persons; an individual may contribute more than one eligible link. Coefficients are post-January-2023 changes in the ages-22–25 versus ages-26–65, beta-Q5 versus beta-Q1 log rate ratio. Entry destination is a log allocation ratio conditional on an observed entry, not an employment-finding or employer-hiring rate. Primary intervals use 9,999 occupation wild-score draws and official longitudinal weights. $MDE_{80} = 2.80\widehat{SE}$ uses the analytic occupation-cluster standard error and is a normal-theory precision diagnostic, not the wild-score rejection threshold and not evidence that a smaller effect is absent.

employment at both endpoints; entry destination conditions on nonemployment-to-employment links and measures allocation across exposure groups. The last is neither a finding probability nor employer hiring. Twelve-month links are endpoints, not sums of monthly transitions. Among fixed-support employed origins, postperiod twelve-month retention ranges from 48.7 to 52.8 percent across young quintiles and from 63.9 to 70.2 percent across older quintiles; preperiod ranges are 50.1–54.0 and 68.8–73.3 percent. Hours (-0.119 , $[-0.295, 0.057]$) and conditional weekly earnings (-0.0245 , $[-0.1024, 0.0534]$) are also imprecise and condition on employment; the legacy earnings series ends in March 2023 and contains only three postperiod months, while a full-window EARNWEEK2 analysis remains unavailable in the authorized extracts.

Observable poststratification does little to the adjacent employment-exit probability contrast: the descriptive Q5–Q1, young–older, post–pre difference is 0.0088 under official link weights and 0.0086 after reweighting retained links to the eligible-origin distribution. At twelve months it moves from 0.0166 to 0.0210. This reweighting requires outcome missingness to be independent of linkage within the observed strata and includes a maximum untrimmed factor of 102.7. Without that assumption, the jointly feasible linear accounting interval for twelve-month employment exit is $[-1.570, 1.575]$, leaving the sign of selection bias unrestricted. These are descriptive probability contrasts and missing-outcome bounds, not the nonlinear regression coefficient and its confidence interval.

The annual primary rule excludes origins whose twelve-month link straddles the January 2023 onset. Coding instead the fraction of destination months exposed gives an employment-exit coefficient of 0.158 ($[-0.070, 0.387]$), compared with the Table 6 primary estimate of 0.123 ($[-0.111, 0.357]$). Both intervals contain zero; no paired interval for their difference is available.

All flow exposure is defined at the origin where an origin occupation exists. No current occupation is imputed to a nonemployed respondent. Annual endpoints miss intervening jobs and transitions, and positive longitudinal-weight eligibility changes the population; neither horizon is a mechanical decomposition of the stock coefficient. Link-rate and observable-balance differences diagnose selection but do not determine its direction, and alternative clustering addresses dependence rather than missing outcomes.

7.3. Payroll bridge and exposure architecture

BCC’s August 2026 19-percent “kept pace” shortfall is a stock-growth comparison, not this regression coefficient or a hiring rate. Complete public membership and tie rules are unavailable, so I use an equal-occupation, tie-preserving approximation and label it a bridge. Its top-two-versus-bottom-three CPS coefficient is -0.0720 ($[-0.1223, -0.0218]$); family-month conditioning gives -0.0168 ($[-0.0808, 0.0473]$). Their paired movement is 0.0553 ($[-0.0013, 0.1118]$), so the design does not detect a difference. Differences in universe, ages, outcome, endpoints, employer controls, and worker–firm flows remain.

That grouping bridge is distinct from changing the regression objective. Giving each of the 468 occupations equal total weight, while retaining WTFINL for within-cell age shares, changes the pooled Q5-by-post coefficient from -0.1321 to -0.0833 ; the paired difference is 0.0488 ($[-0.0604, 0.1581]$). The family-month coefficient changes from -0.0217 to -0.0047 , with paired difference 0.0170 ($[-0.1587, 0.1927]$). These intervals are nondetections, not evidence that the objectives are equivalent.

The bridge therefore asks whether a closer public grouping changes the CPS stock pattern; it does not calibrate CPS transitions to proprietary employer flows. The required entry, exit, job-to-job, occupation-switching, aging, and denominator quantities are not jointly observed.

Finally, $E_1 + \lambda E_2$ varies the weight on tasks requiring complementary software. Here E_1 is the directly accelerable task share and E_2 is the share requiring complementary software. Each row uses rebuilt weights and common draws, and $\lambda = .5$ exactly reproduces beta. Native-tail coefficients range from -0.103 at $\lambda = 0$ to -0.156 at $\lambda = 1$; every comparison with beta includes zero, even though 61.6 percent of stock changes quintile at $\lambda = 0$. This is large classification movement with limited coefficient resolution, not evidence about deployed technology.

Five non- λ definitions give pooled native-support tail coefficients from -0.099 (direct-ability AIOE) to -0.014 (the reversed OECD AI-skills gap). Every alternative-minus-beta paired occupation interval contains zero, although the OECD lower endpoint is -0.0004 .

On the common AIOE support, some family-conditioned implementation contrasts do exclude zero; the result is therefore sensitive to how one construct is mapped in that specification, without implying that the alternatives measure the same technology.

TABLE 7. Weighting direct and software-complement tasks

λ in $E_1 + \lambda E_2$	Q5–Q1 coefficient	Difference from $\lambda = .5$ [95% CI]	Stock changing group (%)
0.00	–0.103	0.029 [–0.044, 0.103]	61.6
0.25	–0.101	0.031 [–0.005, 0.068]	25.5
0.50	–0.132	—	0.0
0.75	–0.143	–0.011 [–0.039, 0.017]	10.2
1.00	–0.156	–0.024 [–0.064, 0.017]	25.5

Notes: All rows use 468 occupations, reconstructed preperiod weights, tie-preserving native groups, Webb software conditioning, and 9,999 common occupation draws. “Stock changing group” is preperiod employment in occupations whose quintile differs from beta ($\lambda = .5$). The coefficient path is descriptive of the capability boundary; no paired beta comparison detects a difference at five percent.

8. Conclusion

Public CPS data contain a sizeable negative association between GPT-4 task exposure and employment for ages 22–25 relative to ages 26–65. The rebuilt Q5–Q1 coefficient is –0.132 log points, but it falls to an imprecise –0.022 when broad occupation families receive their own young-relative monthly paths. Their paired movement is detected; it is a change between conditional estimands, not a causal composition share.

Direct within-family tails cover only 29 occupations and are highly imprecise. Annual ACS data reduce person-sampling uncertainty but do not repair that support: the fixed-label full-calendar family-year coefficient is negative under ACS replicate inference, whereas occupation- and family-shock intervals include zero and the nonreuse-calendar family-year estimate is undetected. Preperiod computer use strengthens rather than attenuates the coefficient, showing that computerization is not settled by one control. Unrestricted dynamics reject joint preperiod equality, and linked-CPS flow estimates do not identify a mechanism or replicate employer-match hiring and separation.

Later self-reported use is descriptively aligned with the score. An exact-code merge to a public RPS workbook retains 121 nonsuppressed occupations: the unweighted exposure–adoption correlation is 0.504 and mean adoption is 0.267 higher in Q5 than Q1. The profile is not monotone, and only 26.2 percent of exposure variation remains after removing broad-family means. Because the rates pool four 2025–26 waves and lack detailed-cell sampling

uncertainty, this check validates neither adoption timing nor an employment mechanism.

The defensible conclusion is therefore comparison-specific. Mapping, support, representation, and conditioning determine what an exposure coefficient measures. A detected movement does not identify why it occurs; nondetection does not establish equivalence. Causal work requires observed adoption or credible assignment and timing variation. This paper instead establishes that the public-CPS young-worker pattern is principally a broad occupational comparison and quantifies what direct within-family and flow evidence cannot resolve. Online Appendix I.3 collects the remaining data, design, and interpretation limits rather than treating them as additional specification searches.

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